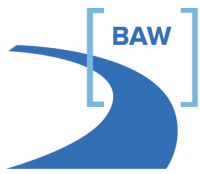


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Kompetenz für die Wasserstraßen

BAW Code of Practice:

Internal Erosion (MMB)

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1 Scope and purpose of the Code

A sufficient level of safety against internal erosion (transport of soil due to hydrodynamic stresses) is a precondition for the stability of waterway structures which are exposed to seepage flows passing through or underneath them. The seepage forces of the water cause mainly the following types of transport of soil particles: contact erosion, piping and suffusion.

The present Code of Practice “Internal Erosion (MMB)” describes verification methods based on the geometric criteria of the soil structure, which are recommended for dealing with specific hydraulic issues (e.g. the design of granular filters according to Code of Practice “MAK (2013)” or verification of internal stability according to the Code of Practice “MSD (2011)”).

2 Terms, definitions and symbols

2.1 Terms and definitions

Revetment

Revetment refers to the complete structure of bank and bottom protection, which includes the armour layer and a filter or the armour and a hydraulic barrier with a separation layer.

Note: As a rule, granular filters are only used for permeable revetments (MAR 2008).

Drain

A drain serves to collect and remove groundwater and seepage water. According to DIN 4095 (1990) the term drain refers both to drain pipes and the drainage layers. For embankments on waterways, drain pipes are not necessarily required.

Filtration stability, mechanical

Mechanical filtration stability is the capability of a filter to retain at a sufficient degree the soil which it is to protect (soil retention capability).

Aggregates

Aggregates are granular material used for construction. Aggregates can be either natural, recycled or industrially manufactured. As a rule, natural aggregates of mineral origin which were only mechanically treated are used for hydraulic engineering purposes (e.g. gravel, sand, crushed rock).

Cohesive / cohesionless soils

The verification of safety against internal erosion requires a distinction between cohesive and cohesionless soils on the basis of the classification according to DIN 18196:2011-05. Cohesive soils as defined by this Code of Practice are fine- and medium-grained soils, which are of at least medium plasticity and have an effective cohesion c' . Cohesionless soils according to this Code of Practice are coarse-grained soils as well as medium- and fine-grained soils of low plasticity.

Granular filters

Granular filters are natural or industrially manufactured aggregate mixes. They can have a single layer design (single-stage filter) or a multiple layer design (multi-stage filter). They must ensure both mechanical and hydraulic filter stability.

2.2 Symbols

Symbol	Designation	Unit
A_{50}	grain size ratio	-
C_C	coefficient of curvature	-
C_U	coefficient of uniformity	-
c_u	undrained shear strength	kN/m ²
c'	effective cohesion	kN/m ²
d_x	grain diameter for x mass % finer by weight	mm
d_D	lower limit of grain-size gap	mm
d_{max}	maximum grain diameter (largest grain)	mm
d_{min}	minimum grain diameter	mm
d_k	relevant pore diameter (Ziems)	mm
d_l	indicative grain diameter according to Lafleur	mm
e	void ratio	-
f	ratio according to Terzaghi	-
F	mass fraction of grains < d (Kenney and Lau)	%
F_s	slip factor	-
h', h''	factors according to Burenkova	-
H	mass fraction between grain diameters d and 4 · d	%
i	hydraulic gradient	-
I_p	plasticity index	-
Index B	index for “base material” (to be retained by the filter)	
Index F	index for “filter”	

3 Basics

3.1 General remarks

Depending on locally acting hydraulic gradients, water seeping through the soil can cause displacements and the transportation of soil particles. These processes can endanger the stability and serviceability of earthworks and solid structures. The following chapters discuss different types of internal erosion caused by hydraulic factors as categorised by Busch et al. (1993), and assess these in terms of their impacts.

Certain geometric conditions in the soil structure are the preconditions for internal erosion to occur. There are corresponding geometric criteria defining limits for the grain diameter and/or the diameter of the pores up to which a particle transport through the pores is geometrically possible.

If geometric conditions allow the movement of particles, a groundwater or seepage water flow is additionally required for internal erosion to occur because the velocity of such flows acts as a sufficiently high force on the individual grain, enabling it to move. The corresponding hydraulic criteria are usually defined based on the limit of the critical hydraulic gradient. When this threshold is exceeded, internal erosion is triggered.

The risk of internal erosion is particularly high in cohesionless soils since there are either no or only negligible cohesive forces between the particles in the soil. The soil particles move freely whenever the pore structure allows it.

In cohesive soils, on the other hand, the soil particles are bound together by chemical and/or physical forces preventing the individual particles from moving alone. However, mobile soil aggregates may form along weak zones in the cohesive soil. Due to the typically prevailing size of soil aggregates, cohesive soils are exposed to a significantly lower risk of internal erosion than cohesionless soils. Since there is a fundamental difference between cohesive and cohesionless soils regarding the verification of safety against internal erosion, the two types of soil are discussed separately in Chapter 4 and Chapter 5.

3.2 Types of internal erosion

3.2.1 Suffusion

Suffusion is the migration and transport of the fine soil fractions through the pores of the granular skeleton of the coarse fractions. The supporting granular skeleton is not altered by these processes, nor is the soil structure destroyed. However, the pore volume and permeability of the soil increase as a result of suffusion, while its density decreases. Due to the higher permeability of the soil the groundwater flow rate increases; however, the hydraulic gradient does not change. Progressive suffusion can facilitate erosion processes if the stability of the supporting granular skeleton is reduced because of the loss of fine soil constituents. If this is the case, the transition from suffusion to erosion can be an ongoing process. As a rule, suffusion is only observed in cohesionless soils.

According to Busch et al. (1993) the following types of suffusion are distinguished depending on where the processes take place: internal suffusion (suffusive erosion processes occur within one layer of the soil), external suffusion (soil is transported towards a free surface) and contact suffusion (soil is transported towards adjacent soil, which is coarser-grained). Internal suffusion is a precondition for all other types of suffusion. Hence, no distinction is made between the different kinds of suffusion; in the following the term suffusion shall refer to internal suffusion as illustrated in Figure 1.

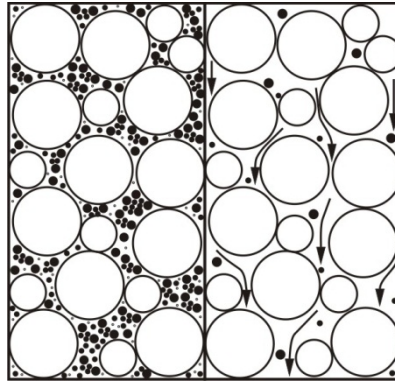


Figure 1: Internal erosion due to internal suffusion

The reverse of the suffusion process is clogging. Clogging is a process where particles transported by the groundwater or seepage water flow are deposited in the existing pores of the granular skeleton of a cohesionless soil. Thus, the pore volume decreases, while the permeability of the soil and the density increase. If moving soil constituents are deposited on the surface of a soil layer, this process is referred to as blinding.

3.2.2 Erosion

Erosion is the migration and transport of almost all grain size fractions of a soil caused by the flow of water. As a consequence, the supporting soil structure changes. Erosion processes can expose earthworks or solid structures to immediate stability risks.

Depending on where erosion occurs, Busch et al. (1993) distinguish between external and internal erosion, as well as between contact erosion and piping at the boundaries of layers or structures.

- External erosion occurs on the free surface of the ground and is usually caused by external loads due to water flow.
- Internal erosion occurs in larger, mostly pipe-like cavities within the ground. Existing cavities (e.g. dead roots, burrows), differences in the bulk density of the soil, soil anisotropy and also prior suffusion processes support the development of internal erosion. If internal erosion develops retrogressively from a free surface of the ground, it can lead to a widening and lengthening of the cavities caused by the concentrated flow (retrogressive erosion), entailing the risk of a collapse due to piping. Retrogressive erosion can be prevented or restrained by using filter layers as a countermeasure to the loss of soil.
- Contact erosion starts at the interface between two soils of different composition (coarse-grained and fine-grained soils) (Figure 2, left): soil particles are transported from the fine-grained soil into the pores of the coarse-grained soil. It is possible for this process to continue as internal suffusion or clogging.
- Piping refers to retrogressive erosion at the interfaces between solid structures and the soil or between a cohesive soil and an underlying cohesionless soil layer (Figure 2, right). Piping leads to the progressive formation of cavities and can cause a collapse.

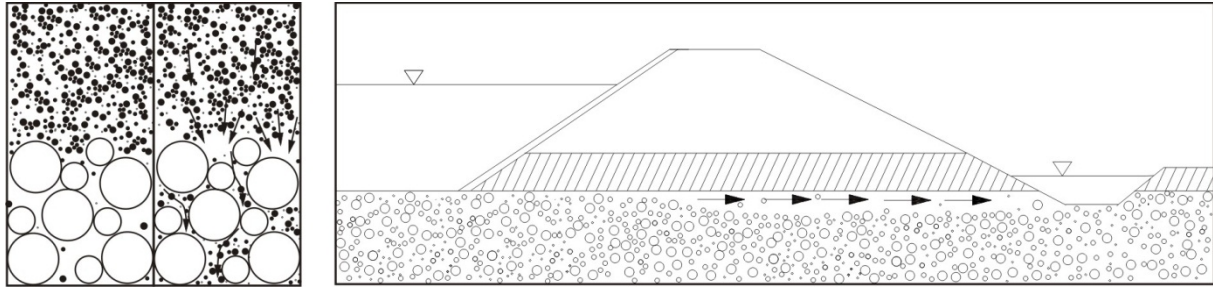


Figure 2: Internal erosion due to contact erosion (left) and piping (right)

3.2.3 Comparison of German and English terms

Since the terminology used here is not standardised in the international literature, the following terms are defined for the purposes of this Code of Practice:

The German term "*Erosion*" ("erosion") is used as the general term for all types of transport of soil particles due to wind or water. If water is the transport medium, the processes either take place on the soil surface or within the granular skeleton. The English term "erosion" is often limited to the phenomenon of surface erosion ("*Oberflächenerosion*"). Erosion in the bed of a waterway is referred to as "scouring". The loss of soil material caused by contact erosion ("*Kontakterosion*") occurring in soil or filter layers below armour layers is also called "winnowing".

The German term "*Innere Erosion*" ("internal erosion") is used as the general term for all transport processes occurring within the soil ("*Suffosion*", "*Kontakterosion*" and "*Piping*" / suffusion, contact erosion and piping). The English term "internal erosion" refers partly to this generic term and partly to suffusion only (in the English literature, the term "suffusion" is used only by a small number of authors). "Internal stability" always refers to suffusive processes.

"*Kontakterosion*" ("contact erosion") is the internal erosion occurring at the interface between two soils of different composition, the more fine-grained fractions being transported through the pores of the more coarse-grained soil. The English literature also uses the term "piping" for this process.

"Piping" exclusively refers to material transport at interfaces.

3.3 Verifications

This Code of Practice exclusively discusses methods for the verification of safety against suffusion and contact erosion on the one hand, and against piping below cohesive soil layers on the other (Figures 1 and 2). The verifications of safety against external erosion occurring on the surface of the ground due to external loads caused by water flows are not covered by this Code of Practice because they are based on hydraulic calculations. Due to the lack of mathematical algorithms for the processes of internal erosion, no verification methods are available either. Whenever damaging effects are possible, preventive measures have to be taken.

For verifications of safety against piping in structures in embankments, a verification method based on the numerical computation of seepage in the area around the structure is recommended in the BAW Code of Practice "Stability of Embankments at German Inland Waterways (MSD 2011)".

The verification of safety against internal erosion according to the present Code of Practice is based exclusively on geometric criteria. Hence the approach is on the safe side, because if the geometric criteria are met, no internal erosion occurs, regardless of the hydraulic loads. Currently there are no generally valid and field-tested verification methods for hydraulic criteria.

4 Verification methods for cohesionless soils

4.1 General remarks

Verifications of safety against internal erosion have to be performed using representative grain size distributions of the soil layers.

For examples of applications of the methods described below refer to the Annexes.

4.2 Suffusion

4.2.1 Verifications of safety against suffusion - general remarks

There is a multitude of methods for the verification of safety against suffusion. They were developed based on different perspectives and classifications (theoretical considerations and/or laboratory tests) and thus differ in terms of their application limits. The recommended methods were selected according to the following criteria:

- broadest possible scope of application,
- simple handling, and
- reliability.

The methods developed by

- Ziems,
- Kenney and Lau, and
- Burenkova,

which are introduced in the following chapters, serve as a basis for the approach recommended in section 4.2.6 for verifications of suffusion resistance.

4.2.2 Simplified method

The verification criteria mentioned below enable an easy and simple assessment of the suffusion risk by reference to the grading curve of the soil. These criteria were derived from Ziems (WAPRO, 1970) and are based on tests carried out on sand and gravel with continuous grading curves.

Soils which meet both of the following criteria are deemed as resistant to suffusion without requiring any particular verification:

- coefficient of uniformity $C_u < 8$ and
- continuous grading curve.

A continuous grading curve is a curve without any grain-size gaps or pronounced changes of curvature.

4.2.3 Method according to Ziems

4.2.3.1 Basics

Ziems' method for verifying safety against suffusion is based on theoretical considerations regarding the pore structure and/or the distribution of constrictions of the pore channels in the soil. These considerations are also fundamental to the filter criteria relating to the interface between two soils (Ziems, 1967).

Ziems equates the constriction size d_k , which is relevant for suffusion, with the mean pore channel diameter according to Pavicic:

$$d_k = 0.455 \cdot \sqrt[6]{C_U} \cdot e \cdot d_{17} \quad (1)$$

The studies on contact erosion conducted by Ziems show that a filter with a relevant constriction size d_k also retains particles smaller than d_k . To allow for these findings, Ziems introduces the slip factor F_s as a reduction factor. This factor takes account of effects such as the grain roughness or non-spherical shapes of natural grains. The slip factor F_s can take values of up to 0.4 (up to 0.6 under dynamic load, e.g. vibration or pulsating flow). Ziems transfers this approach to the verification of suffusion resistance. This is a safe approach because under suffusion conditions the movement of individual grains is more hindered than under contact erosion conditions.

4.2.3.2 Application limits

Determining the mean pore channel diameter according to Pavicic is a valid approach for continuous grading curves only. Hence, the suffusion resistance verification method according to Ziems, which is based on Pavicic's approach, is not suitable for non-continuous grading curves (pronounced changes of curvature or bends, i.e. sudden changes in the slope).

4.2.3.3 Verification method

According to Ziems, a soil is resistant to suffusion if:

$$\frac{d_{\min}}{F_s \cdot d_k} \geq 1.5 \quad (2)$$

with

$d_{\min}$ minimum grain diameter

F_s slip factor (section 4.2.3.1)

d_k relevant constriction size (section 4.2.3.1)

Size $d_{\min}$ defines the largest grain diameter which is transported during suffusion. If $d_{\min}$ corresponds to the grain diameter at 0% passing sieve (d_0) ($d_{\min} = d_0$), no particles will be transported. According to Busch et al. (1993), $d_{\min}$ can be determined as high as d_3 if the corresponding material transport can be tolerated. The factor 1.5 is chosen because the soils are non-uniform regarding their grain size distribution and bulk density.

4.2.4 Method according to Kenney and Lau

4.2.4.1 Basics

Kenney and Lau (1985 and 1986) base their reflections on aspects of the pore geometry of a packing of spheres. They assume that a grain with the diameter d can move through a pore which is formed by several grains with a diameter of $4 \cdot d$. When adopting the verification method according to Kenney and Lau, it is not the entire grading curve which is considered but only the area of the finest fractions, i.e. the maximum portion of the soil which can freely move in the pores of the granular skeleton. Depending on the coefficient of uniformity of the coarse fractions of the soil, $C_{U,coarse}$, these are either the soil's finest-grained 30% where $C_{U,coarse} \leq 3$, or the finest-grained 20% where $C_{U,coarse} > 3$.

Kenney and Lau describe the shape of the grading curve using the relation between H and F , where H is equal to the mass fractions between grain diameters d and $4 \cdot d$, and F is equal to the mass fractions of the grains smaller than d .

To generate the shape curve, for several diameters d of the grading curves the mass fractions F are determined. The $F + H$ mass fractions are also determined, which are related to the diameter $4 \cdot d$. The shape curve is derived from the representation of the H vs. F values in a chart. The straight line $F + H = 1$ is the boundary of the shape curve.

Figure 3 illustrates the definitions of H and F and how the shape curve is constructed.

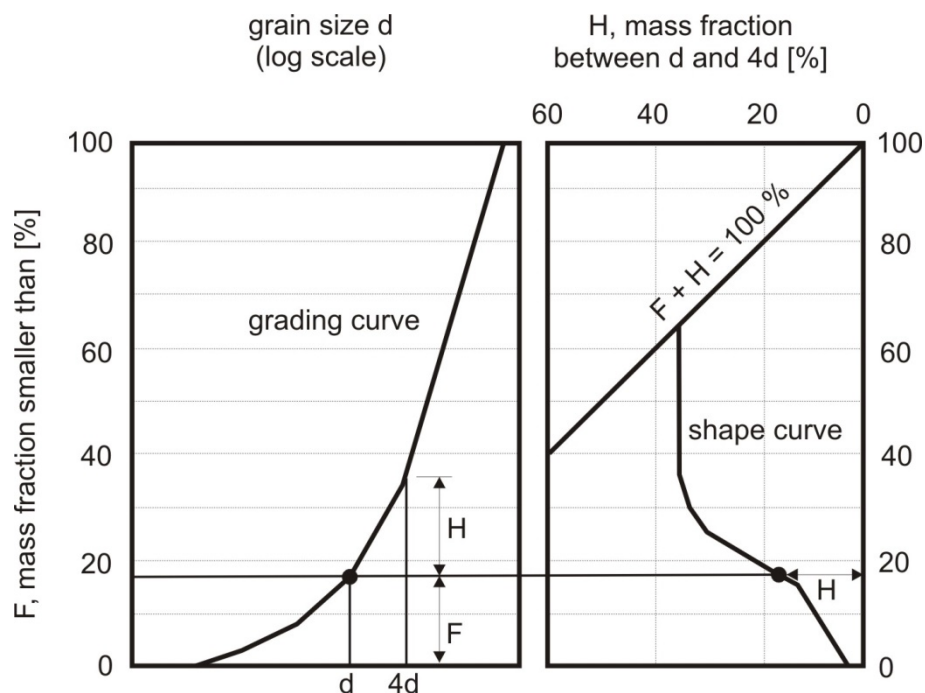


Figure 3: Construction of the shape curve according to Kenney and Lau (1985)

Kenney and Lau verified the method in many tests, which were carried out under unfavourable boundary conditions (vertical seepage flow from the top to the bottom combined with vibration of the sample). This provided a reliable basis for the findings obtained for issues relating to practice.

4.2.4.2 Application limits

For their tests, Kenney and Lau used sands and gravels without any fine-grained fraction ($d < 0.06$ mm). Since the method is based on geometric model assumptions regarding pore geometry, it is equally suitable for more fine-grained, cohesionless soils. Moreover, the method can also be used for soils with gap gradations.

4.2.4.3 Verification method

For the purpose of verifying safety against suffusion the H/F ratio must not be less than a specific value, i.e. the grading curve must have a certain minimum slope.

A distinction is made between a hard criterion with $H = 1.3 F$, and a soft criterion with $H = 1.0 F$. The curve representing $H = 1.0 F$ corresponds to the Fuller curve, i.e. the composition of a graded mineral grain mixture with a minimum share of pores. Both criteria are represented as distinct areas in Figure 4. Laboratory tests conducted by Kenney and Lau have shown that the soft criterion is not always sufficient.

To decide which grain proportion is relevant for verification, the grading curve is split at 30% into fine-grained and coarse-grained fractions. The coefficient of uniformity, $C_{U,coarse}$ is determined for the coarse-grained fractions (70%). If $C_{U,coarse} \leq 3$, the finest-grained 30%, and if $C_{U,coarse} > 3$, the finest-grained 20% are taken into account (Figure 4).

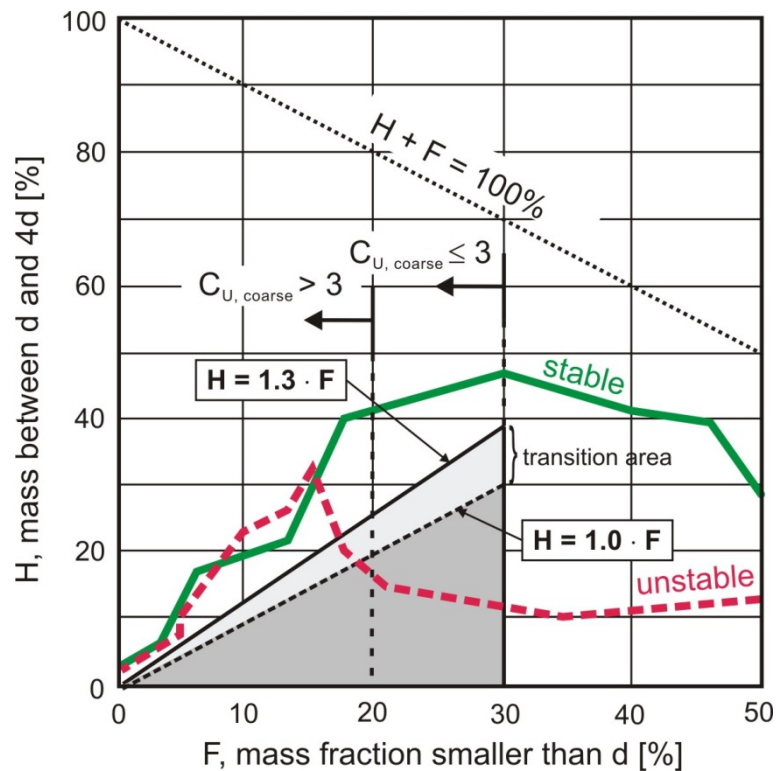


Figure 4: Verification method according to Kenney and Lau (H - F graph)

To conduct a suffusion test, the $H(F)$ curve of the investigated soil material is generated based on the grading curve. It corresponds to the shape curve rotated by 90 degrees (rotation to the right). If the $H(F)$ curve intersects the area shaded in dark grey ($H < F$, soft criterion), the soil is suffusive. If only the light

grey shaded area ($H < 1.3 F$, hard criterion) is intersected by the curve, the soil has to be classified into the transitional area (section 4.2.6). If there are no intersections, the soil has to be deemed as suffusion-resistant. The dashed red line in Figure 4 represents the shape curve of a suffusive soil while the solid green line represents a suffusion-resistant soil.

4.2.5 Method according to Burenkova

4.2.5.1 Basics

The tests conducted by Burenkova (1993) are based on reflections about the structure of a soil, i.e. about its granular skeleton and the filling of the pores with particles moving freely in the soil. Burenkova carried out tests on a large number of soils to determine the grain size at the transition between the granular skeleton and the grains moving freely through the pores.

In order to describe the heterogeneity of a soil, Burenkova uses the representative grain diameters d_{15} , d_{60} and d_{90} and introduces the factors $h' = d_{90} / d_{60}$ and $h'' = d_{90} / d_{15}$.

The evaluation of the tests using the descriptive values h' and h'' yields two boundary curves, which define the area for soils verified as suffusion-resistant in the tests. Figure 5 shows the test results, the boundary curves derived from them and the area (grey-shaded) defined for non-suffusive soils according to Burenkova (1993).

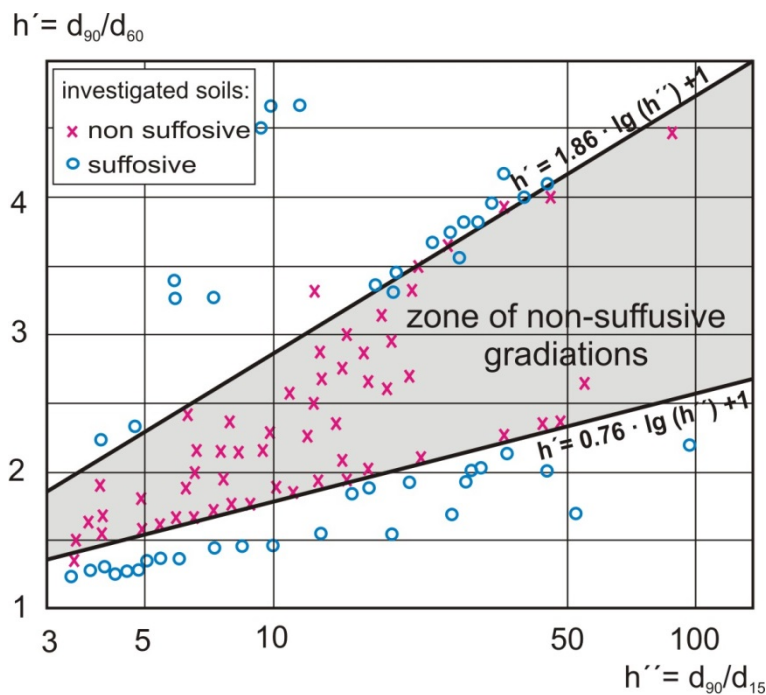


Figure 5: Test results and boundary curves according to Burenkova presented in the h'' - h' chart

4.2.5.2 Application limits

The tests on which the verification method is founded were carried out on soils with a maximum coefficient of uniformity of $C_u = 200$ and a maximum grain diameter of 60 to 100 mm. The grading curves of the investigated soils were concave, convex and linear.

Based on the tests conducted by Burenkova the verification method is applicable to the following soils (Figure 5), where:

- $1 \leq h' \leq 5$ and
- $3 \leq h'' \leq 130$

4.2.5.3 Verification method

According to Burenkova's method a soil is deemed as resistant to suffusion, if:

$$0.76 \cdot \lg(h'') + 1 < h' < 1.86 \cdot \lg(h'') + 1 \quad (3)$$

where $h' = d_{90} / d_{60}$, $h'' = d_{90} / d_{15}$

4.2.6 Verification of safety against suffusion according to MMB

The methods which are discussed here and recommended for application can be distinguished according to their scope of application and complexity. A soil is considered as resistant to suffusion if it is possible to verify safety against suffusion either by applying the simplified criteria, the method according to Ziem's or the method according to Kenney and Lau (in combination with Burenkova's approach, if appropriate). To ensure an approach to verification which is as simple as possible and at the same time suited to a large range of different soil types, the procedure illustrated in Figure 6 is proposed.

In the first step, the simplified criteria (section 4.2.2) are verified. If these criteria are not met, the method pursuant to Ziem's (section 4.2.3) is applied. If the application limits of Ziem's verification method cannot be observed or if the verification process fails to prove safety against suffusion, Kenney's and Lau's method is used (section 4.2.4). For safety reasons, the hard criterion should be applied first (stability criterion $H > 1.3 F$). If this criterion is not met, it must be established whether the soil has to be classified into the transition area between the hard and the soft criterion ($1.0 F < H \leq 1.3 F$). It is advisable in this case to additionally apply Burenkova's method (section 4.2.5). If the shape curve for the soil is in the area of $H \leq 1.0 F$, the soil is suffusive.

However, it is also possible to directly apply the verification method pursuant to Ziem's or Kenney and Lau without previously evaluating the simplified criteria.

The flow chart in Figure 6 illustrates the recommended approach. The dashed line represents the alternative of directly starting with the methods according to Ziem's and/or Kenney and Lau.

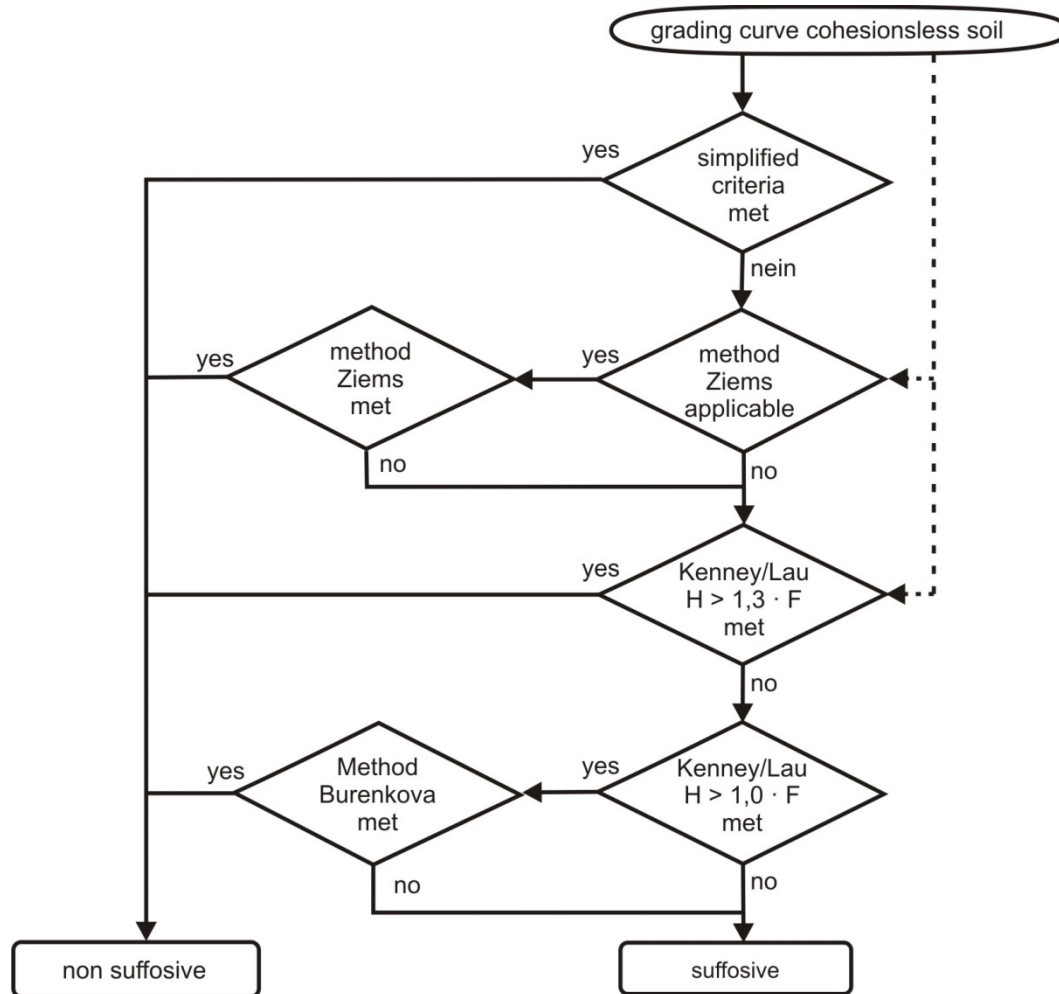


Figure 6: Recommended procedure for verification of safety against suffusion

4.3 Contact erosion

4.3.1 General remarks

Verifications of safety against contact erosion are always required at the interfaces between fine-grained and coarse-grained soils where seepage water flows from the finer to the coarser soil or parallel to the interface.

In the following, the finer soil is designated as base soil (index “B”) and the coarser soil as filter (index “F”) because the verification of safety against contact erosion mostly refers to filter designs and as a rule the corresponding verifications were developed for them.

There are a number of empirically obtained criteria which can be used for verifications of safety against contact erosion (DWA-M 507-1, 2011).

The methods developed by

- Terzaghi,
- Cistin and Ziems,

- Lafleur and
- Myogahara

are recommended for a verification of geometric safety against contact erosion. This chapter provides a description of the fundamentals of each method, the application limits and the verification procedures.

An overview of the application limits of the different methods is provided in Table 1.

Table 1: Application limits of verification methods

Method	$C_{U,F}$	$C_{U,B}$	Filter	Soil	Hydraulic gradient i
Terzaghi	< 2	< 2	narrowly graded sands	narrowly graded sands	< 8
Cistin and Ziems	≤ 18	≤ 20	$d \leq 100 \text{ mm}$	-	≤ 9
Lafleur	< 27	-	sandy gravel $0.06 < d < 50 \text{ mm}$ $d_{15} > 0.2 \text{ mm}$	cohesionless soils $d < 50 \text{ mm}$	≤ 8
Myogahara	-	-	GW, GE or coarser		

The Terzaghi method is subject to many application limits and is only suitable for narrowly graded soils and filters. The method developed by Cistin and Ziems is applicable for soils with a coefficient of uniformity C_U of up to 20, provided there are no grain-size gaps. For very broadly graded soils and soils with grain-size gaps Lafleur's verification method is recommended. The method developed by Myogahara is only suitable for coarse-grained aggregates (e.g. armourstones) used as filters.

All verifications of safety against contact erosion proceed on the assumption that a filter's mechanical stability is sufficient if it is able to retain a specific representative grain diameter of the soil which it is to protect under seepage conditions.

4.3.2 Method developed by Terzaghi and the U.S. Waterways Experiment Station

4.3.2.1 Basics

Terzaghi (1922) conducted investigations and tests relating to the failure of dam foundations due to piping. A conclusion derived from these investigations was that a sufficiently permeable soil surcharge was an "effective remedial measure against collapse due to piping. The soil surcharge must allow the free exit of seepage water while retaining soil particles, i.e. it must assume a filter function". The grain size required for the filter material was determined empirically for the different soil types in this study. A design rule was specified for the first time by Terzaghi and Peck (1948), based on Bertram's laboratory tests (1940) and some first details on filter criteria which were included in previous, unpublished reports by Terzaghi.

According to Terzaghi, a suitable filter material requires that the ratio f of the grain diameter of the filter with 15% finer ($d_{15,F}$) to the grain diameter of the soil to be protected with 85% finer ($d_{85,B}$) is not more than four.

The permissible ratio $f \leq 4$ is conservative, i.e. safe; it is the ratio most widely used in practice. There are other studies which confirm that a ratio $f \leq 5$ (U.S. Waterways Experiment Station, 1941) is sufficient.

4.3.2.2 Application limits

Due to the experimental boundary conditions characterising Terzaghi's criterion, it is only applicable for narrowly graded sands (regarding both the soil and the filter) with $C_U < 2$ and a hydraulic gradient of up to $i = 8$.

4.3.2.3 Verification method

Safety against contact erosion is ensured if

$$f = \frac{d_{15, F}}{d_{85, B}} \leq 5 \quad (4)$$

4.3.3 Method according to Cistin and Ziems

4.3.3.1 Basics

Cistin's paper (1967) describes the phenomena of suffusion in a cohesionless soil and contact suffusion and/or erosion at the interface between two cohesionless soils.

In the tests conducted by Cistin different soils were installed in transparent plexiglas tubes and subjected to a seepage flow from the top, perpendicular to their interface, i.e. in the direction of gravitation. The hydraulic gradient was increased to $i = 9$. The coefficients of uniformity $C_{U,B}$ and $C_{U,F}$ as well as the grain diameters $d_{50,B}$ and $d_{50,F}$ of the test soils were the most important variable soil parameters.

Cistin derived a correlation between the grain size ratio $A_{50} = d_{50,F}/d_{50,B}$ and safety against contact erosion from the test results. Ziems' studies took account, amongst other things, of the influence of the grain shape and the direction of the flow in relation to the interface. The graph drawn up by Cistin and revised by Ziems is shown in Figure 7 (WAPRO, 1970).

4.3.3.2 Application limits

The following boundary conditions are applicable to the method developed by Cistin and Ziems:

1. coefficient of uniformity of the soil: $C_{U,B} \leq 20$,
2. soil has at least a medium bulk density,
3. coefficient of uniformity of the filter: $C_{U,F} \leq 18$,
4. range of grain sizes of the filter $d_i \leq 100$ mm,
5. hydraulic gradient $i \leq 9$,
6. the design assumptions are valid for all flow directions, and
7. the soils must be resistant to suffusion.

4.3.3.3 Verification method

For the purpose of verifying safety against contact erosion the actual grain size ratio A_{50} is derived from the d_{50} values of the material to be analysed (soil $d_{50,B}$, filter $d_{50,F}$) and then compared with the permissible grain size ratio, $\max A_{50}$ as a function of the actual coefficients of uniformity $C_{U,F}$ (filter) and $C_{U,B}$ (soil) as illustrated in Figure 7.

$$A_{50} = \frac{d_{50,F}}{d_{50,B}} \quad (5)$$

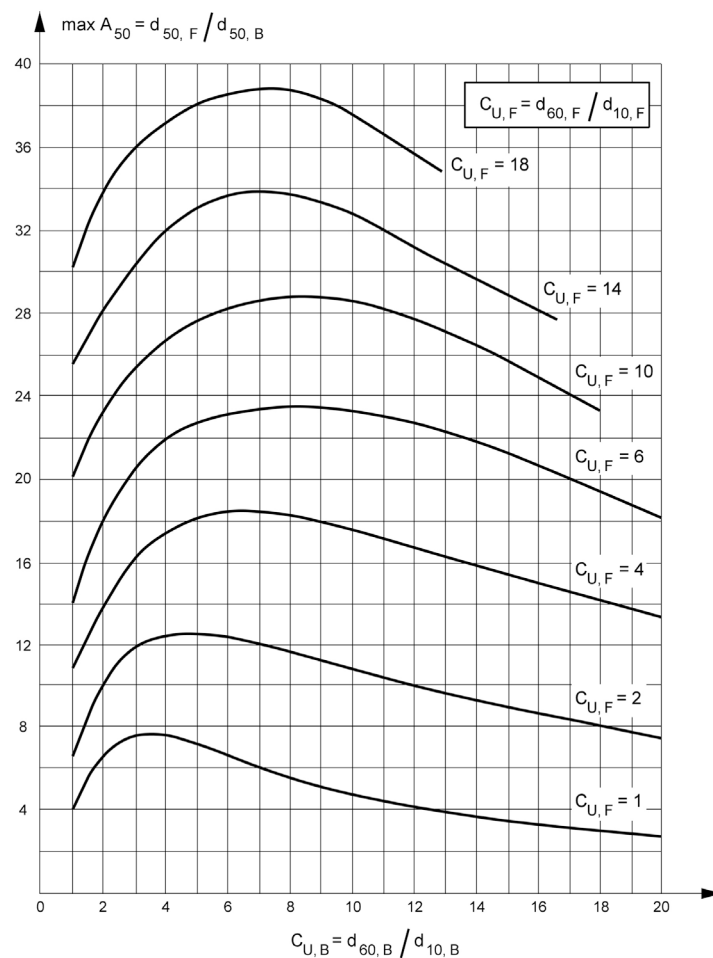


Figure 7: Permissible grain size ratio $\max A_{50}$ according to Cistin and Ziemis

No erosion will occur between the filter and the soil (the soil layers examined) if the actual grain size ratio A_{50} is smaller or equal to the permissible grain size ratio $\max A_{50}$.

$$A_{50} \leq \max A_{50} \quad (6)$$

The graph refers to grains of the coarser soil (filter) which have rounded shapes. With sharp-edged grains the grain size ratio $\max A_{50}$ has to be multiplied by the factor 0.75 (Lattermann, 1997).

4.3.4 Method following the approach by Lafleur

4.3.4.1 Basics

If Lafleur's method is applied, filters can also be dimensioned for non-uniform, i.e. broadly graded or gap-graded cohesionless soils. The method is equally suitable for suffusive soils (Lafleur et al., 1993).

Lafleur conducted tests with a vertical seepage flow using a hydraulic gradient of $i = 8$. The maximum test duration was 71 hours. In addition to his tests, Lafleur also considered the methods and investigations of other authors in his studies (Kenney and Lau, 1984; Sherard, 1984; Bertram, 1940; Terzaghi, 1948).

His reflections are based on the assumption that a certain amount of soil particles will always be washed out at the soil/filter interface and transported into the filter. This results in a complex bridge formation process which will, however, be restricted to the interface area, provided that the soil-filter system is resistant to contact erosion.

In uniform soils the loss of material is very small. In non-uniform soils, a design based on d_{85} (Terzaghi criterion) is not appropriate, as a rule, to ensure a stable base/filter combination.

Lafleur defines a representative ("indicative") grain diameter d_i of the soil for the purposes of filter design. The representative opening width of the filter has to be matched with the grain diameter. Based on his own findings and on studies by other authors (Kenney, Sherard, Bertram), Lafleur defines the representative opening width as $d_{15,F}/5$. Filter stability is ensured if the representative diameter d_i specified for the soil is larger than the filter's representative opening width. Thus it is possible to limit the proportion of washed out soil, and clogging of the filter material can be prevented.

The soil's representative diameter d_i (d_{85} , d_{50} , d_D or d_{30}) depends on the level of the soil's suffusion resistance and the shape of its grading curve in the semi-logarithmic representation. The following basic categories are used for describing the shape of the grading curve:

- linear distribution,
- gap-graded distribution, and
- convex distribution.

Subject to the relevant requirements, these categories are further differentiated (section 4.3.4.2).

4.3.4.2 Application limits

Lafleur's method is based on trials with cohesionless sands and gravels which have a proportion of fine-grained material $d_{0,063}$ of no more than 40% and a hydraulic gradient of up to $i = 8$. The filter material used was sandy gravel ($0.06 < d < 50$ mm, $d_{15} > 0.2$ mm) and the soil material consisted of soils with $d < 50$ mm. The method is only recommended for these soil types and for hydraulic gradients not exceeding $i = 8$.

4.3.4.3 Verification method

In his studies Lafleur deals with a broad range of types of grading curves without providing details on all curve types, however. To ensure the broadest possible application of Lafleur's findings, the verification method discussed below closely follows Lafleur's approach, enlarging it, however, to include an analysis of soils with grain-size gaps.

The recommended verification procedure is illustrated in the flow chart in Figure 8. The starting point is the soil grain size distribution.

Verification of safety against contact erosion using Lafleur's approach begins with an analysis of the soil's suffusion resistance pursuant to section 4.2.

The representative grain diameter d_i of the soil is determined as a function of the shape of the grading curve. According to Lafleur, safety against contact erosion requires that:

$$d_{F,15}/5 < d_i \quad (7)$$

For uniform soils which are resistant to suffusion the verification is the same as according to Terzaghi ($d_i = d_{85}$). Soils with $C_u \leq 6$ and an approximately linear grading curve below d_{90} are deemed uniform soils.

For soils which are characterised by a broadly graded grain size distribution ($C_u > 6$) the shape of the grading curve and the internal stability (suffusion risk) need to be taken into account. In suffusion-resistant soils with a linear, broadly graded grading curve the value of d_i corresponds to the average grain size d_{50} . In suffusion-resistant soils with grain-size gaps, d_i is equal to the lower grain-size of the gap (d_D), however not exceeding d_{50} .

There is a risk in suffusive soils of clogging of the filter surface or the filter itself. A layer of low permeability may consequently form if the filter material is too fine. This may lead to increased pore water pressure. If the filter material is too coarse, this can lead to an excessively high washing out of the fine soil fractions. To take adequate account of both risks, the design of filters to be used in suffusive soils is based on the compromise value of d_{30} to optimise the filter system. In special cases it may be necessary to establish what is more tolerable: increased washing out of soil or higher levels of pore water pressure. Correspondingly, the representative grain diameter d_i would be either higher than d_{30} (with the possible result of an increased loss of soil) or smaller than d_{30} (with the risk of higher pore water pressures). Suffusive soils can have grain-size gaps or clearly convex grading curves. The criterion for gap-graded soils specified in DIN 18196 ($C_u \geq 6$, $C_c = d_{30}^2/(d_{60} \times d_{10}) < 1$ or $C_c > 3$) provides an indication that there are grain-size gaps in the soil.

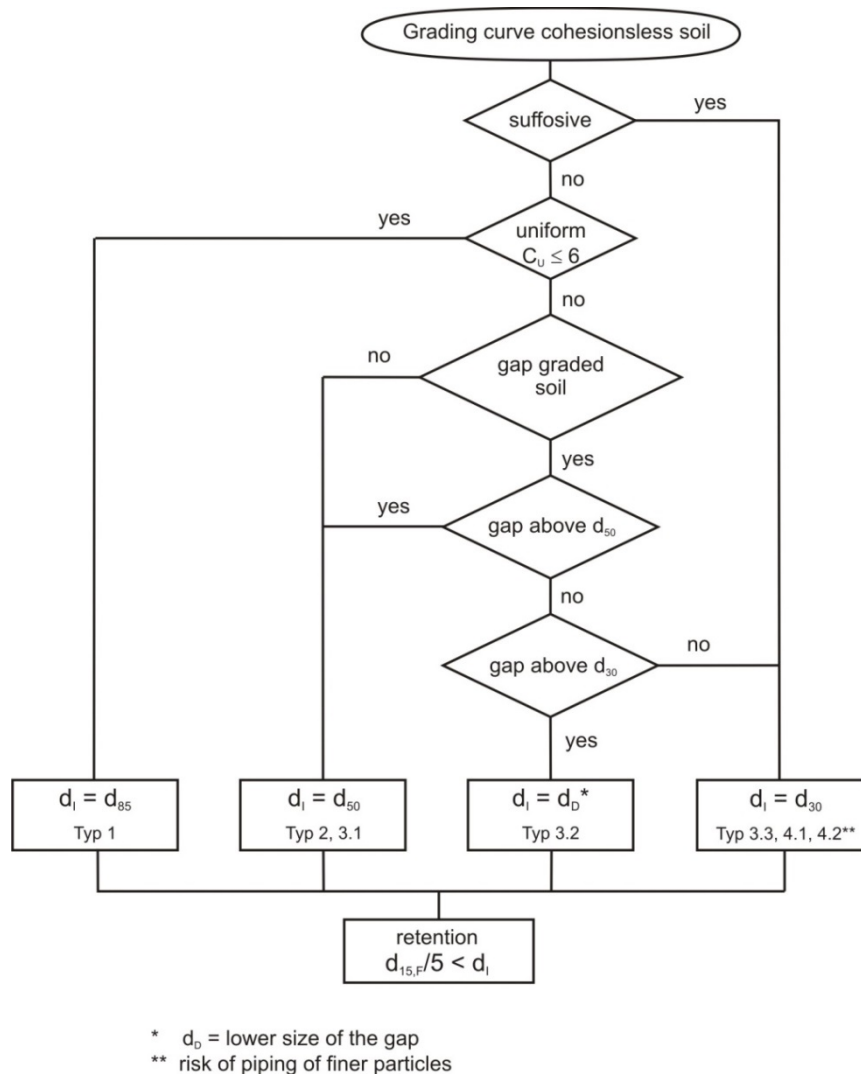


Figure 8: Verification method following Lafleur's approach

The different type curves represented schematically in Figure 9 and Figure 10 provide an orientation for assessing the grading curves to verify safety against contact erosion and are referenced in the flow chart in Figure 8. Figure 9 depicts the type curves of suffusion-resistant soils and Figure 10 depicts the type curves of suffusive soils. The curves do not allow any statement regarding safety against suffusion.

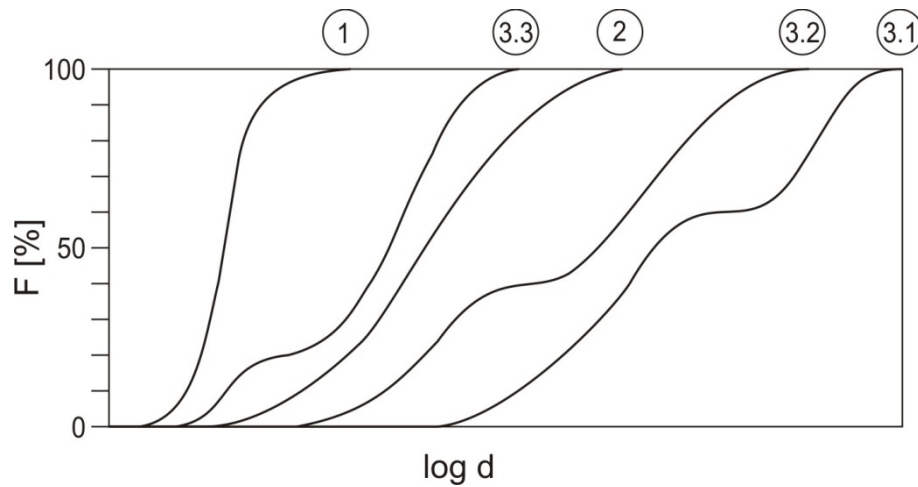


Figure 9: Type curves for suffusion-resistant soils

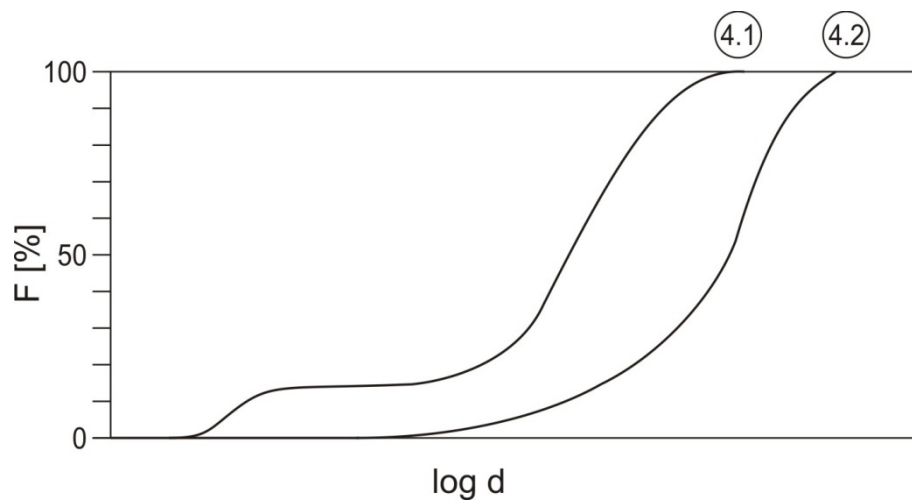


Figure 10: Type curves for suffusive soils

4.3.5 Method according to Myogahara

4.3.5.1 Basics

The method developed by Myogahara (1993) is suitable for verifying safety against contact erosion between coarse fractions, for example between granular filters and armourstones used in revetments. The method takes into account that even if the hydraulic gradients are relatively small (starting at approx. $i = 1$), turbulent flows can occur in the pores of the coarse fractions.

The method was developed on the basis of tests with vertical seepage flows from the top to the bottom. The basic soils analysed for filter stability included 10 different sands and sand-gravel mixtures and a single-grain coarse material with a grain diameter of 53 mm.

4.3.5.2 Application limits

The method is limited to the verification of safety against contact erosion between soils or mineral grain mixtures corresponding to the soil groups GW or GE pursuant to DIN 18196 or of coarser material and a coefficient of uniformity of the base soil $C_{U,B} \geq 8$.

4.3.5.3 Verification method

According to Myogahara, safety against contact erosion requires that:

$$\frac{d_{15,F}}{d_{30,B}} \leq 40 \text{ and } C_{U,B} \geq 8 \quad (8)$$

4.4 Piping

4.4.1 General remarks

At the interface of the percolated ground and a structure or an overlying cohesive soil layer it is possible for cavities (gaps) or loosening zones to form, with no or only low hydraulic resistance. The resulting concentrated flow in these areas can trigger particle transport, causing so-called piping which develops backwards against the flow direction. This is also the case in existing erosion channels formed by dead roots or burrows. The precondition of piping is a transport of the moved particles to a free surface.

4.4.2 Piping in solid structures

The BAW Code of Practice “Stability of Embankments at German Inland Waterways (MSD 2011)”, defines a method for verification of safety against piping in solid structures which takes account of the geometric and geohydraulic conditions in the area of the interface between the ground and the structure. The basis of this verification method is the requirement of sufficient safety levels for the geotechnical and geohydraulic verifications (sufficient safety against hydraulic heave or uplift as well as against failure of the slope) under the most unfavourable hydraulic boundary conditions. To this end, the hydraulic flow calculation assumes the existence of hydraulically effective cavities (gaps) along all interfaces between the structure and the ground, thus precluding any decrease in the hydraulic head, unless the formation of cavities can be excluded due to the characteristics of the ground, the structure’s geometry or the construction method. The distribution of the hydraulic head thus determined is the basis for verifying geotechnical and geohydraulic stability. If the safety requirements are met, there is no risk of soil erosion in the scenario based on the unfavourable assumptions mentioned above. While this is not sufficient to completely prevent particle displacements in the ground, but there is no risk of gradual damage to the embankment due to retrogressive erosion (piping) because no ground material leaves the system.

4.4.3 Piping at interfaces between soil layers

4.4.3.1 Basics

The verification of safety against piping at interfaces between soil layers follows a method developed by Sellmeijer to verify safety against piping beneath a cohesive soil layer (TAW, 1999). The verification method according to TAW is applicable to conditions where the underlying ground is a layer of sand with uniform thickness. Figure 11 depicts the base model of Sellmeijer’s calculation method. The model consists of an impermeable structural component with a length L , which is placed on a sandy ground. The structure is water-loaded on both sides with a water level difference of ΔH between the upstream and the

downstream side. There is an erosion channel with a length l in the area downstream of the contact area. The mathematical verification method is based on the following equations:

- hydraulic head equation for describing the groundwater flow in the sand layer,
- flow equation for the laminar flow in the erosion channel and
- balance of forces equation for the drag force of the seepage flow and the sand grains' resistance to rolling.

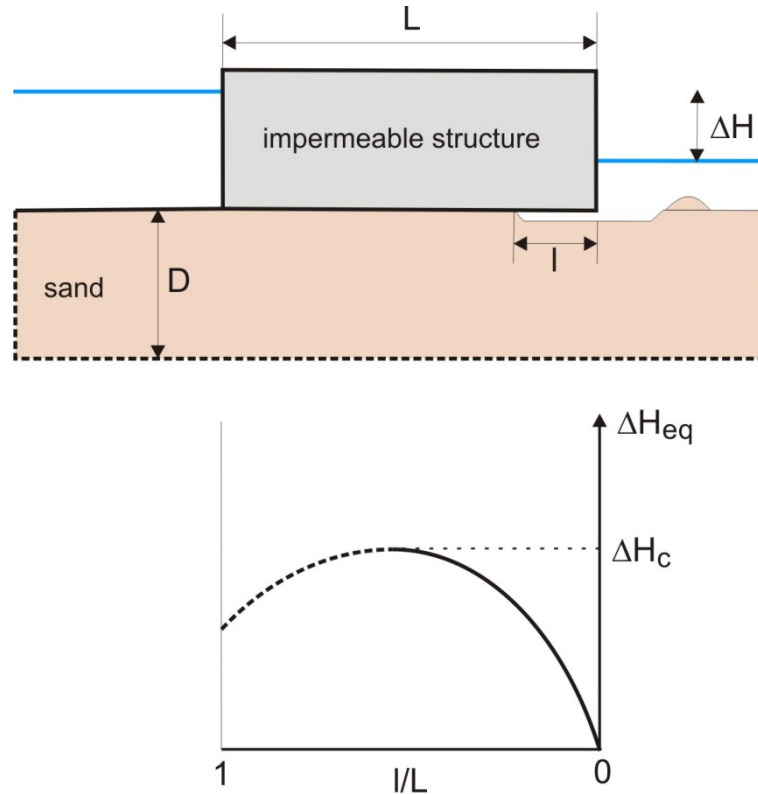


Figure 11: Basic model of the calculation approach according to Sellmeijer (TAW, 1999)

Using these equations it is possible to determine at which maximum water level difference ΔH_{eq} the sand grains are still in equilibrium. This value depends on the ratio of l to L , i.e. the length of the erosion channel l and the length of the structure L ; the drag force coefficient η ; the hydraulic conductivity k of the underlying soil as well as the grain size d_{70} and the angle of rolling resistance θ of the sand grains. According to Sellmeijer's tests conducted for the model shown in Figure 11, the maximum water level difference, i.e. the so-called critical water level difference ΔH_c - the value where an equilibrium state is just still possible - is reached at a ratio of $l/L = 0.5$. If the water level differences ΔH_{eq} are smaller, an equilibrium state will develop and the erosion channel will be restricted, with a length depending on the magnitude of the difference in hydraulic head. When the difference increases, the erosion channel increases in length until a new equilibrium state is reached. As long as the water level difference ΔH_{eq} prevailing in the equilibrium state does not exceed the critical difference ΔH_c , the erosion process is always halted. Once the critical water level difference ΔH_c is reached, the erosion process accelerates and the length l of the erosion channel increases, finally leading to a breakthrough to the upstream side. Based on these model assumptions Sellmeijer performed a multitude of calculations, varying the relevant parameters. To obtain

the analytic equation (9) for the verification, the curves were adjusted to the results of these calculations and validated in large-scale tests.

Sellmeijer transferred his basic approach to the flow situation in Figure 12 which represents an earthwork consisting of cohesive soil located on a layer of cohesionless soil. He found the following equation for determining the critical water level difference ΔH_c .

$$\Delta H_c = \alpha \cdot c \cdot \frac{\gamma_p}{\gamma_w} \cdot \tan \theta \cdot (0.68 - 0.10 \cdot \ln c) \cdot L \quad (9)$$

$$\alpha = \left(\frac{D}{L} \right)^{\frac{0.28}{(D/L)^{2.8} - 1}} \quad (10)$$

$$c = \eta \cdot d_{70} \cdot \left(\frac{1}{\kappa \cdot L} \right)^{\frac{1}{3}} \quad (11)$$

$$\kappa = \frac{\nu}{g} \cdot k = 1.35 \cdot 10^{-7} \cdot k \quad (12)$$

ΔH_c	critical water level difference [m]
γ_p	submerged unit weight of a soil particle [kN/m ³]
γ_w	unit weight of water [kN/m ³]
θ	angle of rolling resistance of the soil [°]
L	length of the seepage path [m]
D	thickness of the layer subjected to seepage [m]
η	drag force coefficient [-]
d_{70}	grain diameter at 70% finer by weight [m]
κ	permeability [m ²]
ν	kinematic viscosity [m ² /s]
k	permeability of the layer subjected to seepage [m/s]
g	acceleration of gravity [m/s ²]

4.4.3.2 Verification method according to TAW

In TAW (1999), a workable formula for the verification of safety against piping at interfaces is derived from Sellmeijer's approach. According to this formula, safety against piping is ensured if the following condition is met:

$$(\Delta H - 0.3 \cdot d) \leq \frac{1}{\gamma} \Delta H_c \quad (13)$$

with the safety factor $\gamma = 1,20$ and "d" as the vertical length of the crack channel in [m].

The critical water level difference ΔH_c applied in equation (13) is the calculation result of equation (9). Table 2 provides recommendations for the selection of parameters. ΔH refers to the actual water level difference and d to the vertical length of the crack channel (Figure 12).

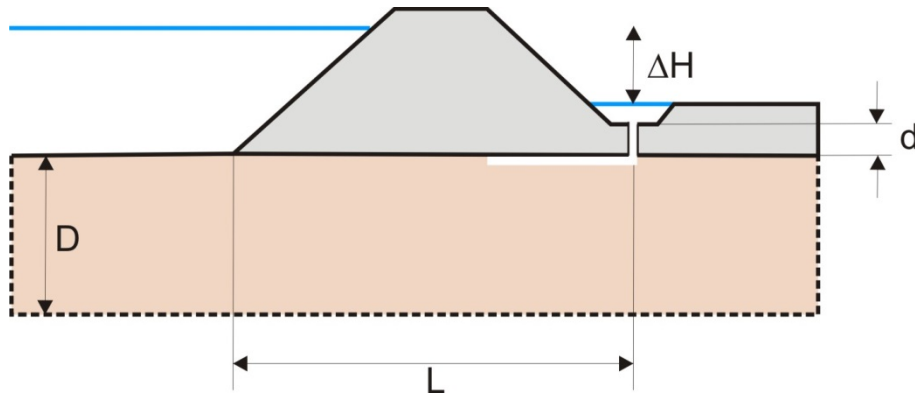


Figure 12: Basic model for verification pursuant to TAW

Where verifications for embankments are concerned, a distinction has to be made whether the top clay layer at the landside embankment toe is continuous ($d > 0$) or either missing or discontinuous ($d = 0$). If there is a continuous top clay layer, the verification method pursuant to TAW (1999) proceeds from the assumption that a crack channel has already formed. Thus it is assumed that failure has already occurred, either as hydraulic heave or a breaking up of the top clay layer. However, pursuant to MSD (2011) safety against hydraulic heave and uplift has to be verified for embankments at Germany's federal waterways. This holds especially true for top clay layers on landside embankment toes if there is a seepage flow under the embankment, e.g. due to the failure of hydraulic barriers. Assuming that the embankment is appropriately protected against hydraulic heave and uplift, piping below the top clay layer is a type of failure impossible to occur and thus needs not to be investigated.

If, however, the top clay layer of an embankment which consists of a cohesive soil material with an underlying layer of cohesionless soil is missing or discontinuous a verification of safety against piping according to TAW (1999) with $d = 0$ is required. Hence, (13) results in:

$$\Delta H \leq \frac{1}{\gamma} \Delta H_c \quad (14)$$

with a safety factor $\gamma = 1.20$ and the parameters according to Table 2.

Table 2: Parameters for verification method according to TAW (1999)

Parameters	Designation	Unit	Type of representative value	Note/ standard deviation V_c
ΔH	water level difference	[m]	mean high water - bottom of ditch mean high water – ground surface	
L	length of the seepage path	[m]	lower characteristic value	if no test values available: $V_c = 0.10$
D	thickness of the layer subjected to seepage	[m]	upper characteristic value	if no test values available: $V_c = 0.10$
d	vertical length of crack channel	[m]	lower characteristic value	
θ	angle of rolling resistance of the soil	[°]	41	
η	drag force coefficient	[-]	0.25	
γ_p	submerged unit weight of a soil particle	[kN/m ³]	17	
γ_w	effective weight density of the water	[kN/m ³]	10	
κ	permeability	[m ²]	upper characteristic value	
d_{70}	grain diameter at 70% passing sieve	[m]	lower characteristic value	0.25

The characteristic values are determined as follows:

- upper characteristic value $\approx \mu (1 + t_{N-1}^{0.95} V_c)$
- lower characteristic value $\approx \mu (1 - t_{N-1}^{0.95} V_c)$

with μ = mean or expected value

V_c = standard deviation

$t_{N-1}^{0.95}$ = Student t-factor (if no test results are available: $t_{N-1}^{0.95} = 1.65$)

5 Verification method for cohesive soils

5.1 General remarks

According to the definition provided in Chapter 2.1, cohesive soils are less susceptible to internal erosion than cohesionless soils. Because of their internal cohesive forces, cohesive soils consist of particles that are fixed in relation to each other and are thus less mobile than the constituents of cohesionless soils. The loads caused by flowing water can cause the liberation of larger soil particles (aggregates) from the bond.

The risk of internal erosion decreases with a higher level of cohesion. At interfaces which are free from tension (e.g. cracks and cavities due to manufacturing defects or natural influences), cohesive soils can

absorb so much water to make them lose almost completely their internal cohesive forces and thus their strength. If seepage flows occur in addition to this, the risk of internal erosion increases as the hydraulic gradient increases.

5.2 Suffusion

As the soil particles in cohesive soils are fixed due to cohesive forces, these soils can be deemed resistant to suffusion without any specific verification being required.

5.3 Contact erosion

5.3.1 General remarks

No verification of safety against contact erosion is required for interfaces between cohesive soils and cohesionless soils because as a rule the occurrence of interfaces which are free from tension (cracks) can be excluded in cohesive soils.

If it is impossible to preclude the formation of cracks in a cohesive soil layer, Sherard's method (1989) should be applied to assess the risk of contact erosion. The formation of cracks in cohesive layers and zones of earthworks is generally possible.

5.3.2 Verification for revetments

For revetments according to the Code of Practice MAR (2008), safety against contact erosion between cohesive and cohesionless soils can be assumed to be sufficient if the conditions specified in Table 3 are met. This applies both to the interface between a cohesive soil and granular filter on the one hand and the interface between a clay liner and cohesionless ground on the other.

Table 3: Verification of safety against contact erosion for cohesive soils in and underneath revetments

Cohesive soil or clay liner	Granular filter or cohesionless soil
$I_p < 0.15$ and $c_u \geq 10 \text{ kN/m}^2$	$d_{10} \leq 0.2 \text{ mm}$ and $d_{60} \leq 0.7 \text{ mm}$
$I_p \geq 0.15$ and $c_u \geq 10 \text{ kN/m}^2$	$d_{10} \leq 0.6 \text{ mm}$ and $d_{60} \leq 2.0 \text{ mm}$

5.3.3 Verification method according to Sherard

5.3.3.1 Basics

The occurrence of damages in embankment dams with an impervious core was the reason why Sherard investigated the filtration properties of low-permeability soils. In each case the damage was caused by contact erosion between the impervious core and the (coarse-grained) shell in places with concentrated leaks. The leaks had in turn been caused by stress redistribution between the impervious core and the shell of the embankment.

In his laboratory tests Sherard examined flow channels with a diameter of 1 to 10 mm in 25 to 100 mm impervious layers (soil) which were exposed to a vertical flow from the top to the bottom with a pressure of approx. 4 bar. Different mixtures of sand, clay and silt were used for the impervious layer without any

differentiation between cohesive and cohesionless soils being made, however. The filter underneath the impervious layer consisted of different types of sand. If the soil contained a proportion of fine-grained material of more than 15% as well as a proportion of gravel with a grain diameter greater than 4.75 mm, the evaluation of the tests only covered the share of the grain size fractions with $d_B \leq 4.75$ mm. In Sherard's studies, the distinction between sand and fine-grained material is based on the grain diameter of $d_B \leq 0.074$ mm as the threshold value.

Sherard found in his tests that during seepage particles of the impervious layer are liberated from the flow channel's wall, suspended, and finally enter the filter (sand).

If the filter material is sufficiently fine, the suspended soil particles settle in the granular skeleton and form a stable filter cake with low permeability. This halts the loss of soil.

If the filter is too coarse, no stable filter cake will form. The transport of suspended material to the sand is not halted; the flow channel becomes larger, leading to the failure of the impervious core.

Based on the test results, Sherard defines various criteria for an assessment of contact erosion between the soil and the filter. To this end, he classifies the soil into four types according to their proportion of grains with $d_B < 0.074$ mm: one cohesive soil (soil type 1 in Table 4), two soils which can be either cohesive or cohesionless (soil types 2 and 3 in Table 4) and one cohesionless soil. The cohesionless soil type is not included in the following discussion.

The relevant design size for the filter is the diameter $d_{15,F}$.

5.3.3.2 Application limits

Although Sherard also studied cohesionless soils, his method is only applied to cohesive soils in this Code of Practice. It should only be used if there is a risk of cracks forming in the cohesive layer.

5.3.3.3 Verification method

In a first step, it has to be established whether the soil is a cohesive soil as defined in Chapter 2.1. Any grain size fractions of $d_B > 4.75$ mm have to be excluded from all further calculations. For the verification method described below only the finer-grained material ($d_{100,B} = 4.75$ mm) is considered. Based on the proportion of soil particles with $d_B \leq 0.074$ mm, the soil is assigned to one of the three soil types listed in Table 4. The design size for the filter $d_{15,F}$ must be determined and the risk of contact erosion assessed.

Table 4: Soil types and verification criteria according to Sherard

Soil type	Soil description	Criterion
soil type 1	$d_{85,B} < 0.074$ mm	$d_{15,F} \leq 9 \cdot d_{85,B}$ and $d_{15,F} > 0.2$ mm
soil type 2	$d_{40,B} < 0.074$ mm $d_{85,B} \geq 0.074$ mm	$d_{15,F} = 0.7$ mm
soil type 3	$d_{15,B} < 0.074$ mm $d_{40,B} \geq 0.074$ mm	$d_{15,F} \leq \frac{40 - F_{0.074}}{40 - 15} \cdot (4 \cdot d_{85} - 0.7) + 0.7 \text{ [mm]}$ $F_{0.074}$ is the proportion of soil with $d_B < 0.074$ mm without any grains with $d > 4.75$ mm

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MAR (2008): Merkblatt Anwendung von Regelbauweisen für Böschungs- und Sohlensicherungen an Wasserstraßen (MAR) (Code of Practice “Use of Standard Construction Methods for Bank and Bottom Protection on Inland Waterways”), Federal Waterways Engineering and Research Institute (BAW), Karlsruhe

TAW (1999): Technical Report on Sand Boils (Piping), Technical Advisory Committee on Flood Defences (TAW), The Netherlands

WAPRO (1970): Werkstandard Nachweis der Beständigkeit von Erdstoffen gegenüber der Einwirkung der Sickerwasserströmung, Suffosion nichtbindiger Erdstoffe, WAPRO 4.04 Blatt 2; VEB Projektierung Wasserwirtschaft, Halle (WAPRO standard for the verification of the resistance of soils to seepage water flows, suffusion of cohesionless soils, WAPRO 4.04 Sheet 2; VEB Projektierung Wasserwirtschaft, Halle)

Annex 1: Examples of verifications of safety against suffusion according to MMB (section 4.2)

The verification of safety against suffusion has to be performed for each grading curve located within a size range.

For the verification of safety against suffusion for the grading curve depicted in Figure A1.1 (SU* according to DIN 18196) the method described in section 4.2.6 is applied.

A void ratio of 0.5 was determined.

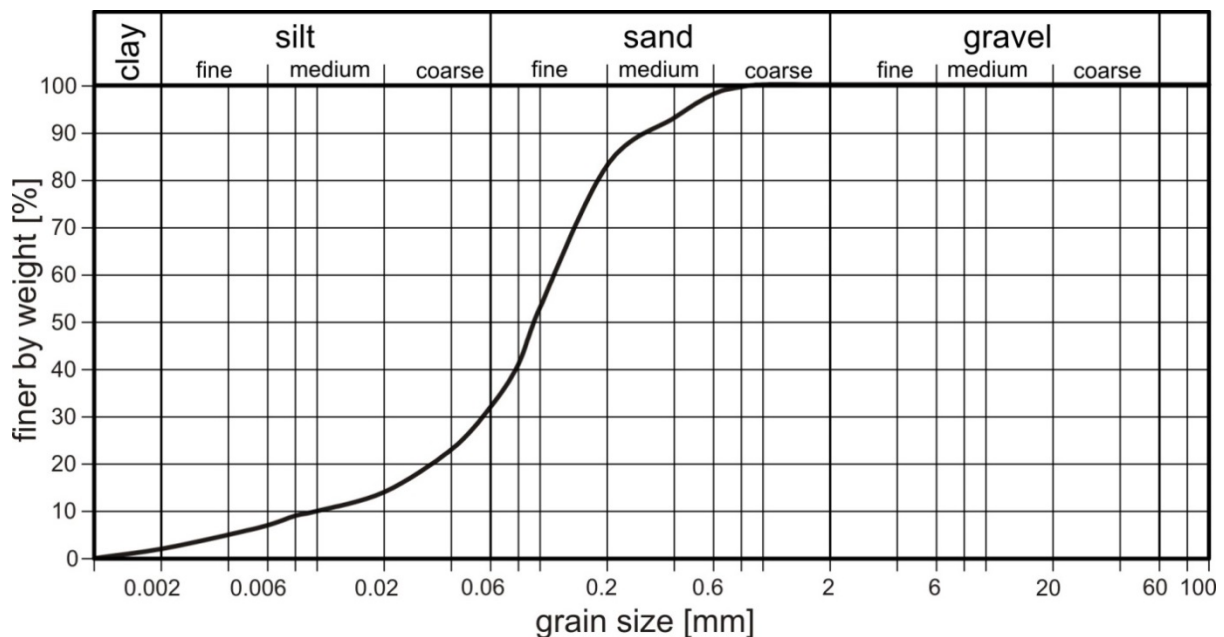


Figure A1.1: Grading curve

Step 1: Examination regarding cohesiveness (section 2.1)

Since there is a fundamental difference between cohesive and cohesionless soils, the first step must be an analysis of the soil to establish whether it is cohesive or cohesionless.

Where no detailed laboratory tests are available, the distinction is made according to the definition provided in section 2.1, based on the soil classification pursuant to DIN 18196. Accordingly, soils belonging to the groups UL, TL and coarser soils are categorised as cohesionless.

The soil represented by Figure A1.1 is classified as SU* according to DIN 18196, i.e. it is a cohesionless soil type.

Step 2: Verification adopting the simplified method (section 4.2.2)

Coefficient of uniformity: $C_U = d_{60} / d_{10} = 0.12 \text{ mm} / 0.01 \text{ mm} = 12$

$C_U > 8$

→ The simplified method is not appropriate.

Step 3: Verification according to the Ziem's method (section 4.2.3)

No grain-size gaps, no pronounced changes of the curvature, no bends

→ The method developed by Ziem's may be applied.

According to Ziem's, a soil is resistant to suffusion if:

$$\frac{d_{\min}}{F_S \cdot d_k} \geq 1.5$$

$d_{\min} = d_3 = 0.003 \text{ mm}$ (chosen parameter: 3% material loss is tolerated)

$F_S = 0.4$ (max F_S assuming a steady state flow)

$$d_k = 0.455 \cdot \sqrt[5]{C_U} \cdot e \cdot d_{17} = 0.455 \cdot \sqrt[5]{12} \cdot 0.5 \cdot 0.023 \text{ mm} = 0.0079 \text{ mm}$$

$$\frac{0.003 \text{ mm}}{0.4 \cdot 0.0079 \text{ mm}} = 0.95 < 1.5$$

According to Ziem's method the soil is suffusive. As specified by the flow chart in Figure 6 (section 4.2.6), safety against suffusion has to be further verified by the Kenney and Lau method.

Step 4: Verification according to the Kenney and Lau method (section 4.2.4)

a) Identification of the fine-grained area to be examined

Instead of examining the complete grading curve, Kenney and Lau look only at the finest-grained 20 or 30 mass percent. To this end, the grading curve is split up at 30 mass percent (Figure A1.2). If the coefficient of uniformity is $C_{U, \text{coarse}} \leq 3$ for the coarse-grained fractions, the finest 30% are considered in the verification. If $C_{U, \text{coarse}} > 3$, the finest 20% are relevant for verification (Figure A1.4).

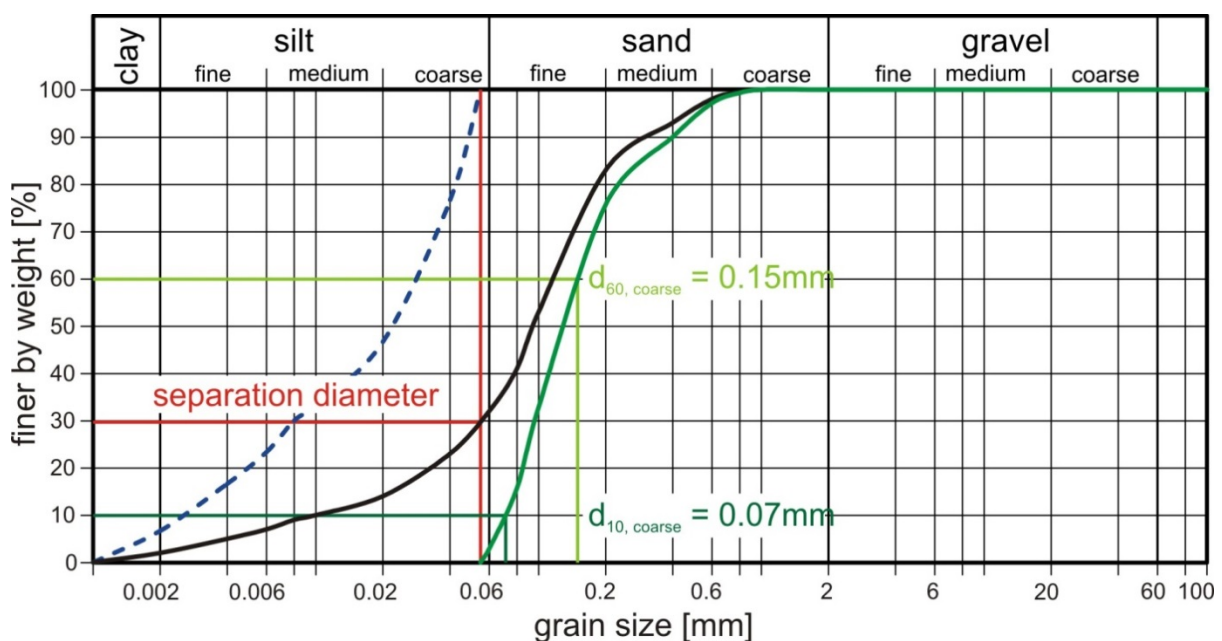


Figure A1.2: Identification of $C_{U, \text{coarse}}$; splitting up at 30%

Splitting up of the finest 30% (Figure A1.2).

$$C_{U, \text{coarse}} = d_{60, \text{coarse}} / d_{10, \text{coarse}} = 0.15 \text{ mm} / 0.07 \text{ mm} = 2.1 < 3$$

Since $C_{U, \text{coarse}} \leq 3$, the finest 30% have to be considered (Figure A1.4).

b) Determination of the H(F)-curve based on the grading curve

The grading curve is transformed into an H(F) curve for the purposes of the verification.

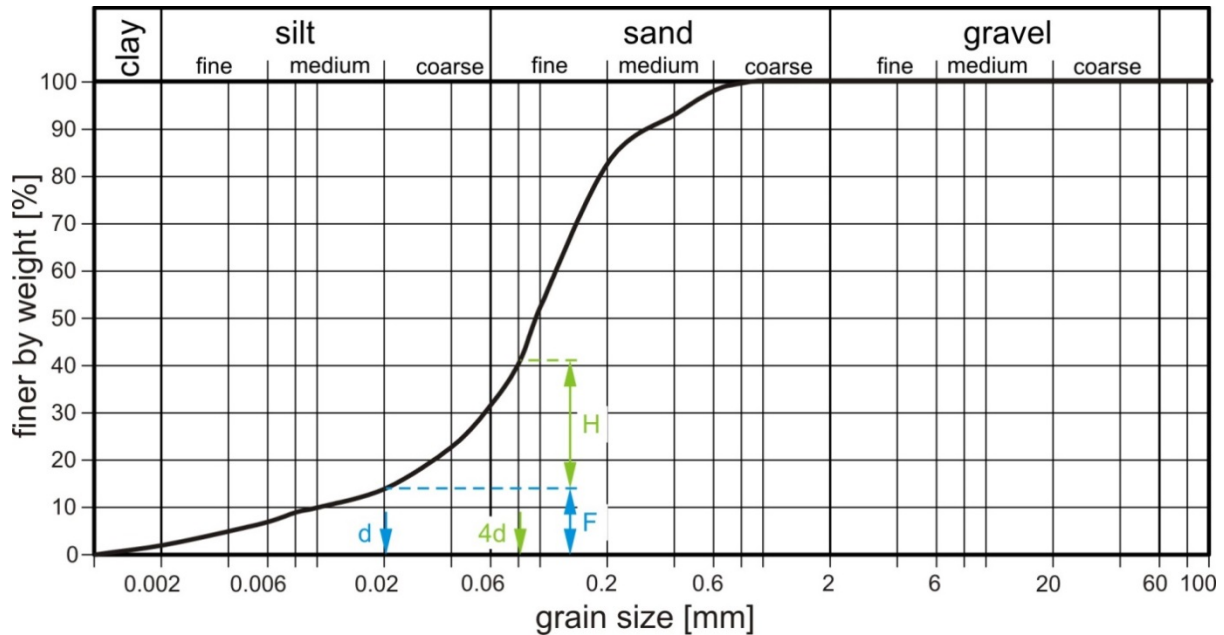


Figure A1.3: Parameters for determining the H(F) curve

The basic approach is demonstrated using the example of a diameter of $d = 0.02$.

$$4 \cdot d = 4 \cdot 0.02 \text{ mm} = 0.08 \text{ mm}$$

$$F = 14\% \quad (\text{mass proportion at } d, \text{ refer also to Figure 2})$$

$$H = 41\% - 14\% = 27\% \quad (\text{mass proportion at } 4d \text{ minus } F, \text{ refer also to Figure 2})$$

If this approach is repeated for several diameters between d_0 and d_{30} or d_{20} , the H(F) curve depicted in Figure A1.4 is obtained. *(Note: In the present example, determining the H(F) curve for up to $F = 30\%$ ($d = d_{30}$), is sufficient for a verification of safety against suffusion.)*

If the H(F) curve is located above the straight line $H = 1.3 \cdot F$ (for $F \leq 30\%$), the soil has to be considered as resistant to suffusion.

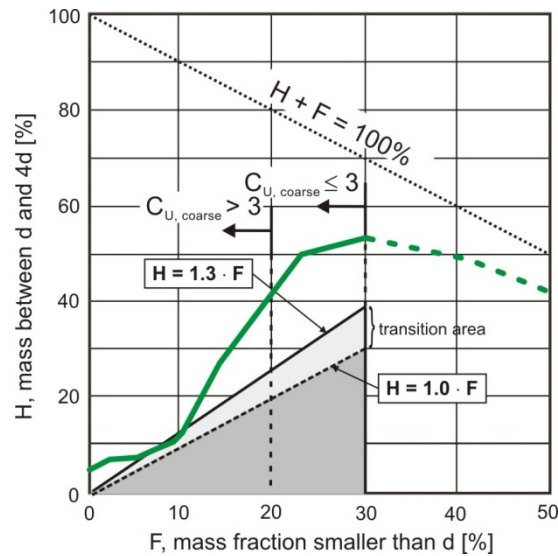


Figure A1.4: $H(F)$ graph according to Kenney and Lau

Pursuant to the Kenney and Lau method, the soil type must be assigned to the transition zone between the hard criterion ($H = 1.3 \cdot F$) and the soft criterion ($H = 1.0 \cdot F$). Hence, the additional verification according to Burenkova is required (Figure 6, section 4.2.6).

Step 5: Verification according to Burenkova's method (section 4.2.5)

According to Burenkova's method a soil is deemed to be resistant to suffusion if:

$$0.76 \cdot \lg(h'') + 1 < h' < 1.86 \cdot \lg(h'') + 1$$

Where

$$h' = d_{90} / d_{60} = 0.30 \text{ mm} / 0.12 \text{ mm} = 2.5 \quad \text{and}$$

$$h'' = d_{90} / d_{15} = 0.30 \text{ mm} / 0.021 \text{ mm} = 14.3$$

the result is

$$0.76 \cdot \lg(14.3) + 1 = 1.88$$

$$1.86 \cdot \lg(14.3) + 1 = 3.15$$

$$1.88 < 2.50 < 3.15$$

According to Burenkova's method the soil is resistant to suffusion.

Step 6: Evaluation of the findings

Since according to Kenney and Lau the soil has to be classified into the transition zone and since Burenkova's method establishes its resistance to suffusion, the soil is finally assessed to be resistant to suffusion.

Annex 2: Examples of verifications of safety against contact erosion according to MMB (section 4.2)

In the following, the finer soil 1 is designated as base soil (index “B”) and the coarser soil 2 as filter (index “F”).

Example 1: Verification according to Terzaghi’s method (section 4.3.2)

Figure A2.1 shows the grading curves of the soils used for the following calculation examples.

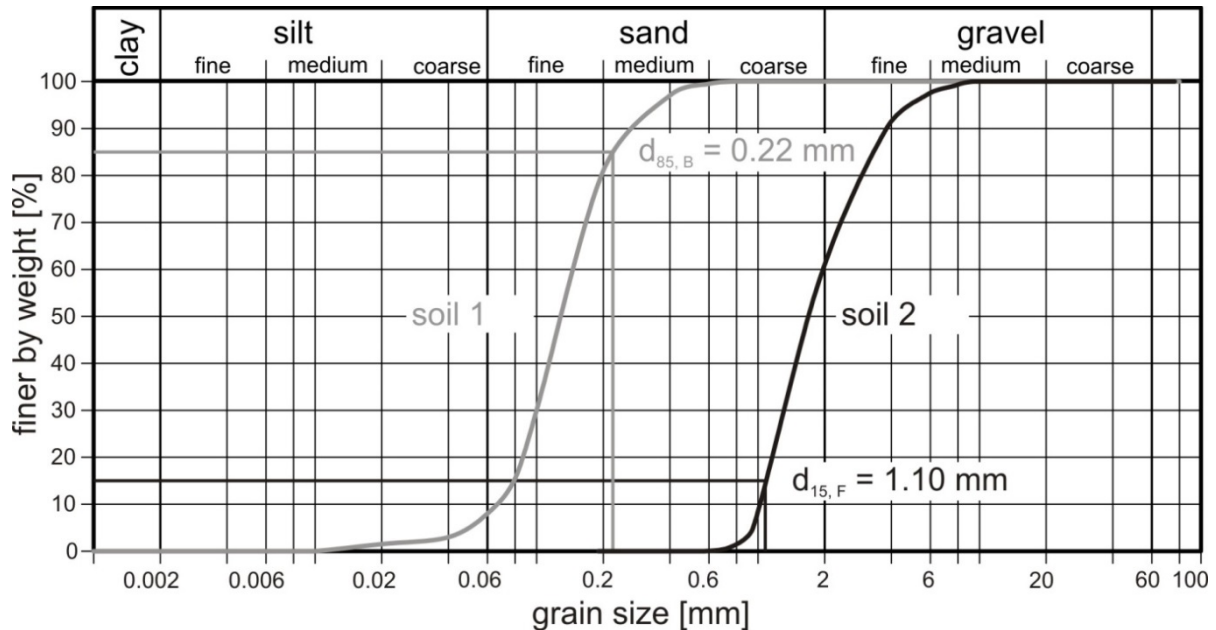


Figure A2.1: Grading curves

Step 1: Assessing the application limits

Terzaghi’s method is applicable only to sands with a coefficient of uniformity of $C_u < 2$.

$$C_{u,B} = d_{60,B} / d_{10,B} = 0.13 \text{ mm} / 0.066 \text{ mm} = 1.97$$

$$C_{u,F} = d_{60,F} / d_{10,F} = 2.0 \text{ mm} / 1.05 \text{ mm} = 1.90$$

The method is suitable for the soils represented in Figure A2.1.

Step 2: Verification method

According to Terzaghi, no contact erosion between two soils occurs if:

$$f = \frac{d_{15,F}}{d_{85,B}} \leq 5$$

$$d_{15,F} = 1.10 \text{ mm}, d_{85,B} = 0.22 \text{ mm} \quad (\text{Figure A2.1})$$

$$f = \frac{1.1 \text{ mm}}{0.22 \text{ mm}} = 5 \leq 5$$

Based on this requirement according to Terzaghi, the soils are resistant to contact erosion.

Example 2: Verification according to the Cistin-Ziems method (section 4.3.3)

Figure A2.2 shows the grading curves of the soils used for the following calculation examples.

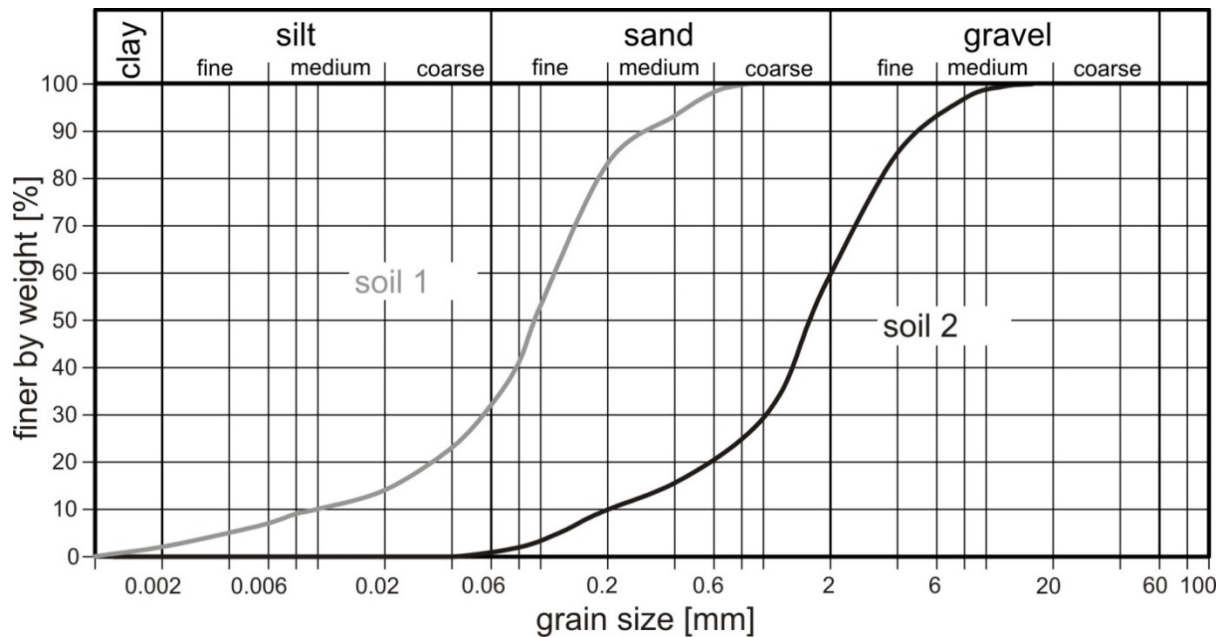


Figure A2.2: Grading curves

Step 1: Assessing the application limits (section 4.3.3.2)

$$d_{10,B} = 0.01 \text{ mm}$$

$$d_{60,B} = 0.12 \text{ mm}$$

$$C_{U,B} = d_{60,B} / d_{10,B} = 0.12 \text{ mm} / 0.01 \text{ mm} = 12$$

1. $C_{U,B} = 12 \leq 20$ condition met

$$d_{10,F} = 0.2 \text{ mm}$$

$$d_{60,F} = 20 \text{ mm}$$

$$C_{U,F} = d_{60,F} / d_{10,F} = 20 \text{ mm} / 0.2 \text{ mm} = 10$$

2. $C_{U,F} = 10 \leq 18$ condition met
3. $d_{\max,F} = 20 \text{ mm} \leq 100 \text{ mm}$ condition met
4. assumption $i \leq 9$ is fulfilled
5. not relevant
6. The application of Cistin-Ziems method requires the soil to be resistant to suffusion.

The verification is performed pursuant to the methods described in section 4.2.

Soil 1 ("B"):

Suffusion resistance has been demonstrated in Annex 1; soil 1 is resistant to suffusion.

Soil 2 („F“)**a) Simplified method**

$C_{U,F} > 8$: the use of the simplified method is precluded

b) Verification according to Ziems

No grain-size gaps, no pronounced curvature, no bends

→ The method developed by Ziems can be applied for soil 2.

$$d_{\min,F} = d_3 = 0.08 \text{ mm} \quad (3\% \text{ material loss tolerated})$$

$$F_S = 0.4 \quad (\text{steady state flow})$$

$$e = 0.5 \quad (\text{test result})$$

$$d_k = 0.455 \cdot \sqrt[6]{C_U} \cdot e \cdot d_{17}$$

$$d_k = 0.455 \cdot \sqrt[6]{10} \cdot 0.5 \cdot 0.4 \text{ mm} = 0.13 \text{ mm}$$

$$\frac{d_{\min}}{F_S \cdot d_k} \geq 1.5$$

$$\frac{0.08 \text{ mm}}{0.4 \cdot 0.13 \text{ mm}} = 1.54 \geq 1.5$$

→ Soil 2 is resistant to suffusion.

Step 2: Verification method

For the verification of safety against contact erosion according to Cistin and Ziems the actual grain size ratio A_{50} is compared with the permissible grain size ratio, $\max A_{50}$. No erosion will occur between the soil layers examined if the actual grain size ratio A_{50} is smaller than or equal to the permissible grain size ratio $\max A_{50}$.

Parameters required for the finer-grained soil 1:

$$d_{10,B} = 0.01 \text{ mm}$$

$$d_{60,B} = 0.12 \text{ mm}$$

$$C_{U,B} = d_{60,B} / d_{10,B} = 0.12 \text{ mm} / 0.01 \text{ mm} = 12$$

$$d_{50,B} = 0.09 \text{ mm}$$

Parameters required for the coarser-grained soil 2:

$$d_{10,F} = 0.2 \text{ mm}$$

$$d_{60,F} = 20 \text{ mm}$$

$$C_{U,F} = d_{60,F} / d_{10,F} = 20 \text{ mm} / 0.2 \text{ mm} = 10$$

$$d_{50,F} = 1.6 \text{ mm}$$

Assumption: grains of the coarser soil have rounded shapes.

$$\max A_{50} = 27.6 \quad (\text{value obtained from Figure 7, section 4.3.3.3})$$

$$A_{50} = d_{50,F} / d_{50,B} = 1.6 \text{ mm} / 0.09 \text{ mm} = 17.8 < \max A_{50}$$

No erosion can occur between the soil layers!

Example 3: Verification following the approach by Lafleur (section 4.3.4)

Figure A2.3 shows the grading curves of the soils used for the following calculation examples. The grading curves depicted fail to meet the requirements regarding the application limits of the methods of Terzaghi and Ziems. Consequently, the verification method following Lafleur's approach has to be applied.

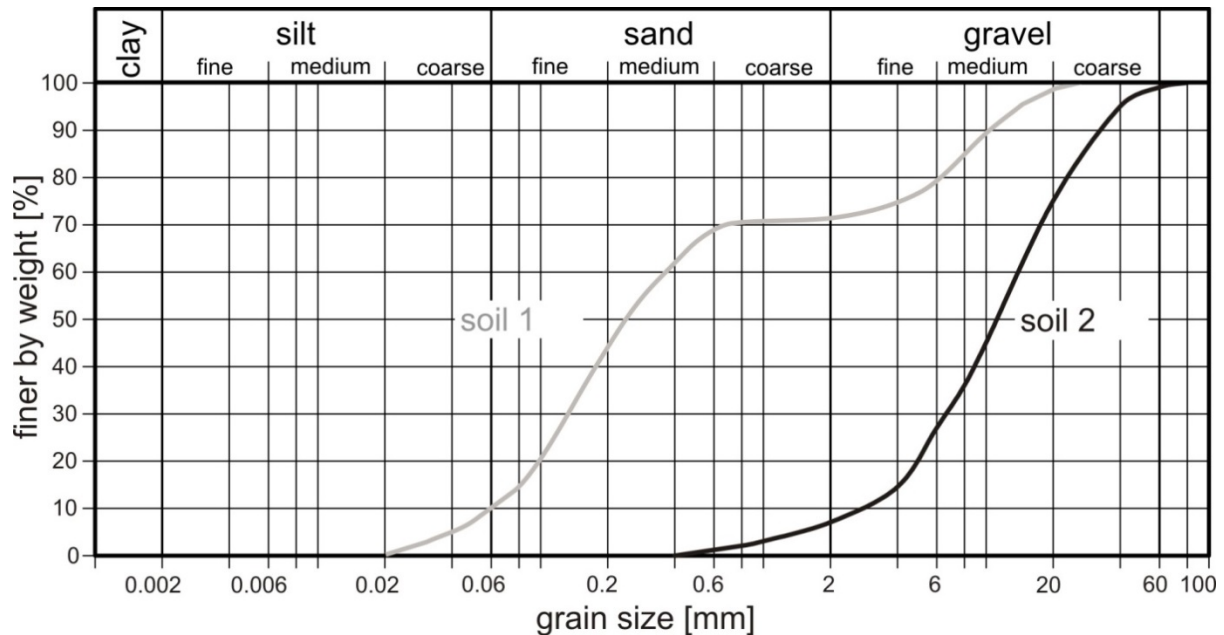


Figure A2.3: Grading curves

The verification is performed in accordance with the flow chart (Figure 6, section 4.3.4.3).

Step 1: Examination of soil 1 regarding resistance to suffusion

The verification is performed pursuant to the methods described in section 4.2.

Since soil 1 has no continuous grading curve neither the simplified method nor the method developed by Ziems is suitable for the assessment of suffusion resistance.

Verification according to Kenney and Lau:

$$C_{U, \text{coarse}} = d_{60} / d_{10} = 2.4 \text{ mm} / 0.16 \text{ mm} = 15 > 3$$

(Note: the splitting up of the grading curve at 30% is not shown here)

Since $C_{U, \text{coarse}} > 3$, only the finest 20% have to be considered (Figure A1.4).

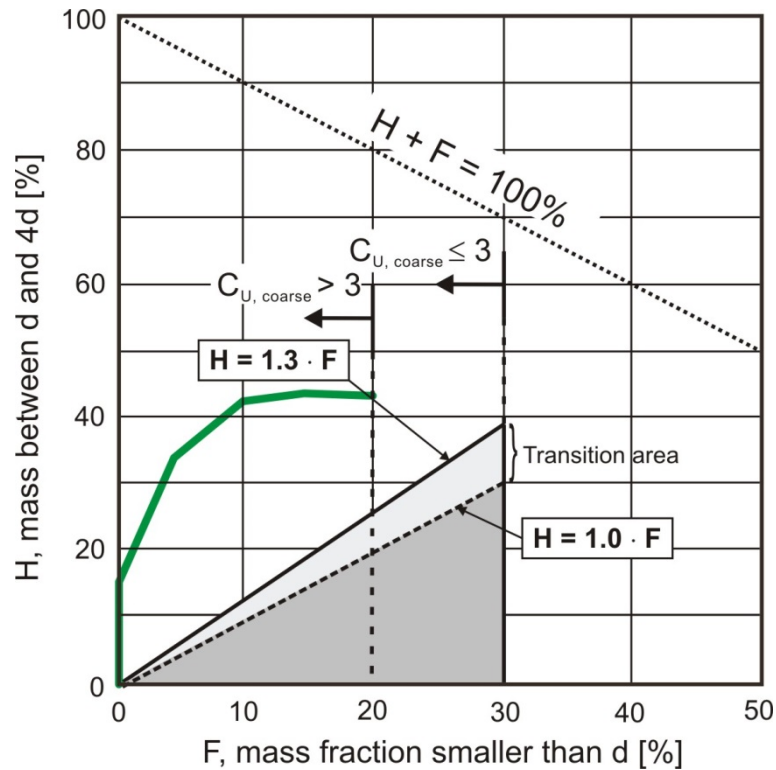


Figure A2.4: $H(F)$ graph according to Kenney and Lau

Verification according to the Kenney and Lau method demonstrates that the soil is resistant to suffusion because the $H(F)$ curve is constantly above the boundary line $H = 1.3 \cdot F$ of the hard criterion.

Step 2: Examination of the soil regarding uniformity

The coefficient of uniformity of soil 1 is

$C_{U,B} = d_{60,B} / d_{10,B} = 0.37 \text{ mm} / 0.06 \text{ mm} = 6.2 > 6$. The soil is a non-uniform soil.

Step 3: Examination of the soil for grain-size gaps

In the range of coarse sand the soil shows a gap above $d_{50,B}$.

Step 4: Verification

According to Lafleur's method safety against contact erosion is established if:

$$d_{15,F} / 5 < d_i$$

After completion of the verifications under steps one to three, soil 1 is assigned to soil type 3.1 according to section 4.3.4.3.

For type 3.1, $d_i = d_{50,B}$ (Figure 7, section 4.3.4.3)

$$d_{50,B} = 0.23 \text{ mm}$$

$$d_{15,F} / 5 = 4.1 \text{ mm} / 5 = 0.82 \text{ mm} > 0.23 \text{ mm}$$

Regarding geometric criteria, the soil layers are **not resistant to contact erosion** in relation to each other.

Annex 3: Example of verification of safety against piping according to MMB (section 4.4)

The example chosen (Figure A3.1) is an embankment with low permeability located on a sand layer (SW) with a thickness of 8 m.

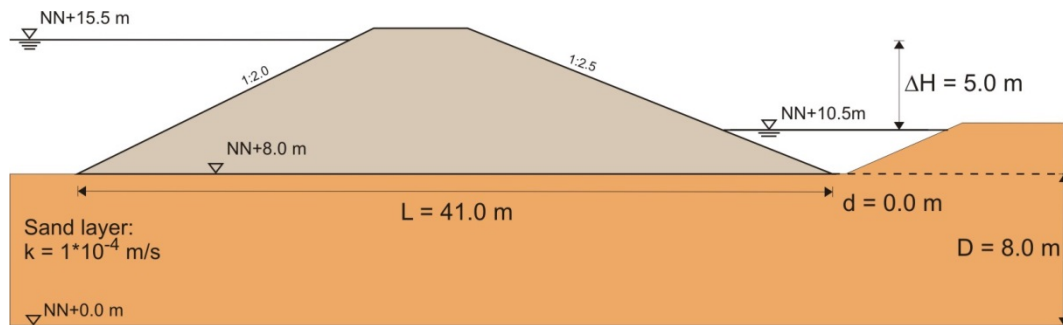


Figure A3.1: Embankment cross-section for verification of safety against piping

Safety against piping is ensured according to section 4.4.3 provided the following condition is met:

$$\Delta H \leq \frac{1}{\gamma} \Delta H_c$$

Table A3.1 provides an overview of the parameters needed to determine ΔH_c , which are derived from Table 2 and formulas (9) to (12) in section 4.4.3.2.

Table A3.1 Input parameters for verification of safety against piping

Symbol	Value	Unit	Designation
d_{70}	0.001	[m]	grain diameter for 70% passing sieve
θ	41	[°]	angle of rolling resistance of the soil
γ_p	17	[kN/m³]	submerged unit weight of a soil particle
γ_w	10	[kN/m³]	unit weight of water
L	15	[m]	length of seepage path
D	8	[m]	thickness of the layer subjected to seepage
d	0	[m]	length of crack channel
η	0.25	[-]	drag force coefficient
ν	$1.3 \cdot 10^{-6}$	[m²/s]	kinematic viscosity
g	9.81	[m/s²]	acceleration of gravitational
γ	1.2	[-]	safety factor

The values specified in Table A3.1 and the formulas used in section 4.4 enable the critical water level difference ΔH_c to be determined as follows.

$$\alpha = \left(\frac{D}{L} \right)^{\left(\frac{0.28}{(D/L)^{2.8} - 1} \right)} = \left(\frac{8 \text{ m}}{41 \text{ m}} \right)^{\left(\frac{0.28}{(8 \text{ m}/41 \text{ m})^{2.8} - 1} \right)} = 1.59$$

$$\kappa = \frac{\nu}{g} \cdot k = \frac{1.3 \cdot 10^{-6} \text{ m}^2 / \text{s}}{9.81 \text{ m} / \text{s}^2} \cdot 1 \cdot 10^{-4} \text{ m} / \text{s} = 1.36 \cdot 10^{-11} \text{ m}^2$$

$$c = \eta \cdot d_{70} \cdot \left(\frac{1}{\kappa \cdot L} \right)^{\frac{1}{3}} = 0.25 \cdot 0.001 \text{ m} \cdot \left(\frac{1}{1.36 \cdot 10^{-11} \text{ m}^2 \cdot 41 \text{ m}} \right)^{\frac{1}{3}} = 0.30$$

$$\Delta H_c = \alpha \cdot c \cdot \frac{\gamma_p}{\gamma_w} \cdot \tan \theta \cdot (0.68 - 0.10 \cdot \ln c) \cdot L$$

$$\Delta H_c = 1.59 \cdot 0.30 \cdot \frac{17 \text{ kN} / \text{m}^3}{10 \text{ kN} / \text{m}^3} \cdot \tan 41^\circ \cdot (0.68 - 0.1 \cdot \ln 0.30) \cdot 41 \text{ m} = 23.35 \text{ m}$$

$$\Delta H = (15.5 \text{ m} - 5.5 \text{ m}) = 5.0 \text{ m}$$

$$\frac{1}{\gamma} \cdot \Delta H_c = \frac{1}{1.2} \cdot 23.35 \text{ m} = 19.46 \text{ m}$$

$$(\Delta H - 0.3 \cdot d) = 5.0 \text{ m} \leq 19.46 \text{ m} = \frac{1}{\gamma} \Delta H_c$$

→ No continuous erosion channel from the landside to the waterside is formed; consequently, there is no risk of a collapse due to piping.

Note: In addition to the verification of safety against internal erosion (i.e. piping erosion in this example), the Code of Practice MSD requires a verification of safety against collapse due to hydraulic heave at the toe of the embankment.